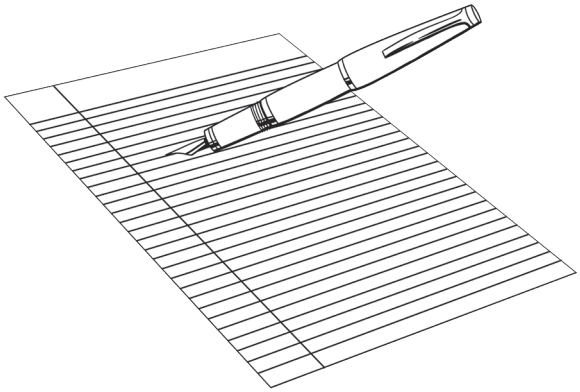
Natural Science

Term 3 Exam Revision Paper

Grade 8 **Total: /100**

Name: _____ Class: _____

Date: _____ **Time: 1½ Hours**



Instructions

- **Read all the questions carefully.**
- **Answer all questions.**
- **Show all calculations where necessary.**
- **Write neatly and legibly.**

Section A

Question 1: Multiple Choice **20**

Choose the correct answer and write only the letter (A, B, C or D) next to the question number.

1.1 What is the main cause of static electricity?

A. Moving magnets

B. Rubbing two materials together

C. Heating a metal

D. Mixing chemicals

1.2 Which of the following materials is a good conductor of electricity?

A. Rubber

B. Plastic

C. Wood

D. Copper

visit twinkl.co.za

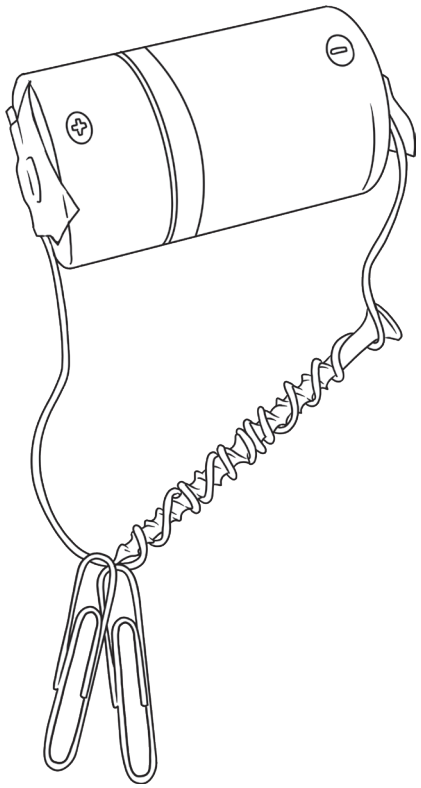
1.3 What effect does an electric current have when it passes through a conductor?

A. It cools the conductor

B. It heats the conductor

C. It changes the colour of the conductor

D. It makes the conductor invisible



1.4 What is an electromagnet?

A. A magnet that uses chemical energy

B. A magnet that uses electrical energy

C. A magnet made from a piece of wood

D. A permanent magnet

1.5 What is the main component of a simple electromagnet?

A. A plastic rod

B. An iron nail

C. A piece of string

D. A piece of cloth

1.6 During electrolysis, what is produced at the positive electrode?

A. Anions

B. Cations

C. Electrons

D. Protons

1.7 If one component fails in a series circuit, what happens to the other components?

A. They continue to work

B. They become brighter

C. They stop working

D. They explode

visit twinkl.co.za

1.8 What is the main advantage of a parallel circuit?

A. Components are cheaper

B. Components last longer

C. Each component can be controlled independently

D. It uses less electricity

1.9 Which safety device is used to prevent overloading in an electrical circuit?

A. A switch

B. A fuse

C. A resistor

D. An ammeter

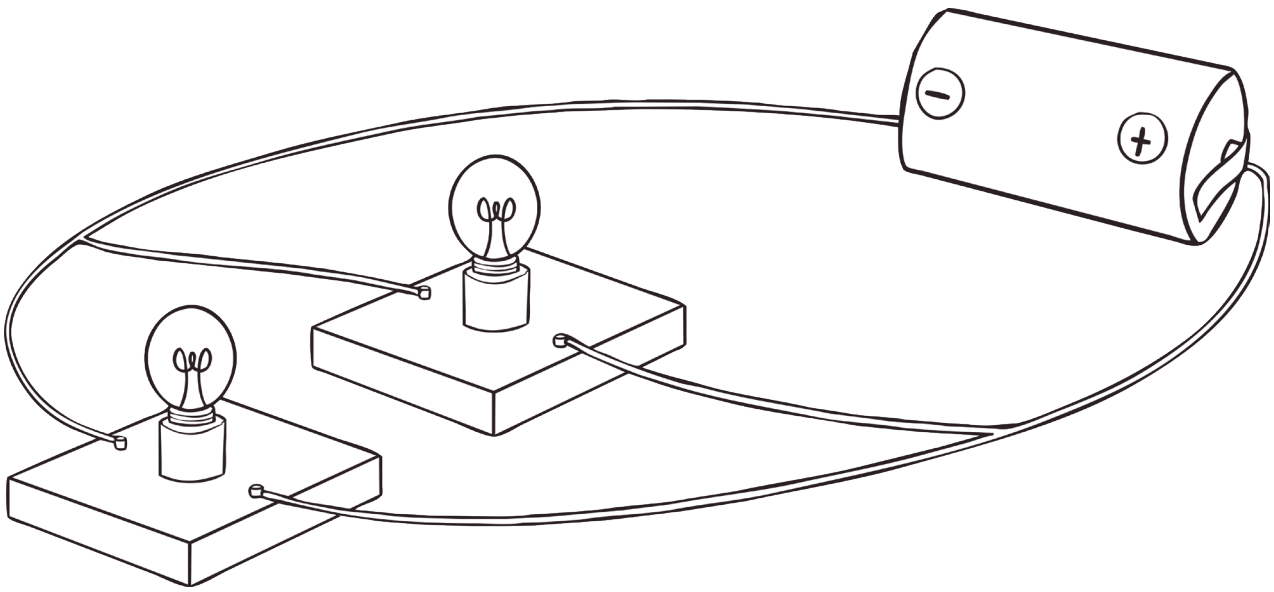
1.10 Which of the following is a chemical effect of an electric current?

A. Heating a wire

B. Lighting a bulb

C. Electroplating

D. Turning a motor



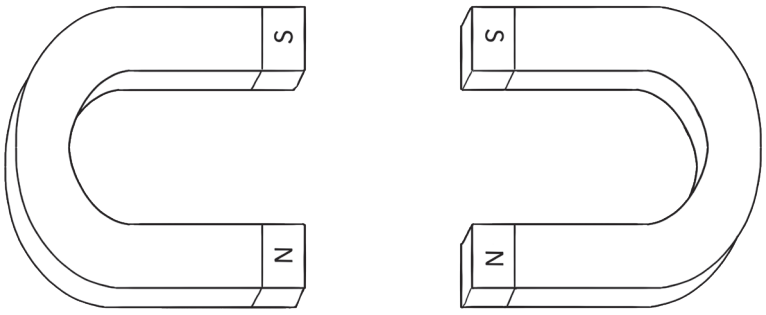
visit twinkl.co.za

Question 2: Column A and Column B Matching

10

Match the terms in Column A with their correct descriptions in Column B. Write only the letter (A, B, C, etc.) next to the number in Column A.

Column A	Answer	Column B
2.1. Static electricity		A. Allows electric current to pass through it easily
2.2. Conductor		B. Electric current splits water into hydrogen and oxygen
2.3. Series circuit		C. Uses electricity to create a magnetic field
2.4. Electromagnet		D. An electrical circuit with components connected end-to-end
2.5. Electrolysis		E. A safety device that melts when the current is too high
2.6. Parallel circuit		F. The flow of electric charge due to the rubbing of two materials
2.7. Magnetism		G. An electrical circuit with multiple paths for the current
2.8. Heating effect		H. Production of heat when an electric current passes through a conductor
2.9. Safety fuse		I. The force exerted by magnets when they attract or repel each other
2.10. Chemical effect		J. Changing the chemical composition of a substance by passing an electric current through it



visit twinkl.co.za

Section B



Question 3: Short Answer Questions

30

3.1. **Static Electricity**

a. Explain what happens when you rub a balloon against your hair. (3)

b. Why do objects become charged during this process? (2)

3.2. **Effects of Electric Current**

a. Describe the heating effect of an electric current with an example. (2)

b. Explain the concept of an electromagnet and one of its practical uses. (3)

3.3. **Magnetism and Electromagnetism**

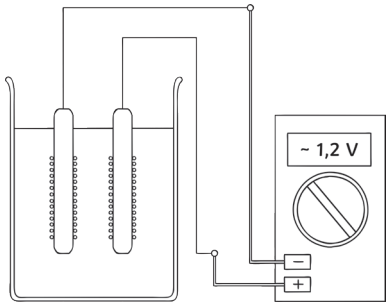
a. What is the difference between a permanent magnet and an electromagnet? (4)

visit twinkl.co.za

b. Describe how you can make a simple electromagnet. (4)

3.4. **Electrolysis**

a. Define electrolysis and give one example of its application. (2)



b. What happens to ions during the process of electrolysis? (2)

3.5. **Series and Parallel Circuits**

a. Draw a simple series circuit with three bulbs and one cell. Label all the parts. (4)

visit twinkl.co.za

b. Draw a simple parallel circuit with three bulbs and one cell. Label all the parts. (4)

Section C

Question 4: Long Answer Questions — 20

4.1. **Safety with Electricity**

a. Explain the importance of safety measures when working with electricity. Provide three examples of safety measures. (4)

b. Describe the function of a fuse and how it protects electrical appliances. (4)

visit twinkl.co.za

4.2. **Practical Application of Electromagnetism**

a. Discuss how electromagnets are used in everyday life. Provide at least two examples. (4)

b. Explain the relationship between electricity and magnetism in the context of electromagnets. (4)

4.3. **Chemical Effects of Electric Current**

a. Describe the process of electroplating and its benefits. (4)

Total Marks: 70

visit twinkl.co.za

Natural Science

Term 3 Exam Memorandum

Section A

Question 1: Multiple Choice — 20

1.1 **B** Rubbing two materials together

1.2 **D** Copper

1.3 **B** It heats the conductors

1.4 **B** A magnet that uses electrical energy

1.5 **B** An iron nail

1.6 **A** Anions.

1.7 **C** They stop working

1.8 **C** Each component can be controlled independently

1.9 **B** A fuse

1.10 **C** Electroplating

Question 2: Column A and Column B Matching — 10

2.1 **F** The flow of electric charge due to the rubbing of two materials

2.2 **A** Allows electric current to pass through it easily

2.3 **D** An electrical circuit with components connected end-to-end

2.4 **C** Uses electricity to create a magnetic field

2.5 **B** Electric current splits water into hydrogen and oxygen

2.6 **G** An electrical circuit with multiple paths for the current

2.7 **I** The force exerted by magnets when they attract or repel each other

2.8 **H** Production of heat when an electric current passes through a conductor

2.9 **E** A safety device that melts when the current is too high

2.10 **J** Changing the chemical composition of a substance by passing an electric current through it

Section B

Question 3: Short Answer Questions — **30**

3.1. **Static Electricity**

a. Explain what happens when you rub a balloon against your hair. (3)

When you rub a balloon against your hair, electrons are transferred from your hair to the balloon. ✓ This causes your hair to become positively charged ✓ and the balloon to become negatively charged. ✓

b. Why do objects become charged during this process? (2)

Objects become charged during this process due to the transfer of electrons from one object to another, ✓ creating an imbalance of charges. ✓

3.2. **Effects of Electric Current**

a. Describe the heating effect of an electric current with an example. (2)

The heating effect of an electric current can be seen in devices like electric heaters and toasters. ✓ When an electric current passes through a high-resistance wire, it produces heat ✓.

b. Explain the concept of an electromagnet and one of its practical uses. (3)

An electromagnet is created when an electric current flows through a coil of wire wrapped around an iron core. ✓ This generates a magnetic field.✓ One practical use is in electric doorbells, ✓ where the electromagnet moves a striker to hit the bell.

3.3. **Magnetism and Electromagnetism**

a. What is the difference between a permanent magnet and an electromagnet? (4)

A permanent magnet is a material that maintains its magnetic properties ✓ even without an external electric field. ✓ An electromagnet, however, is a type of magnet whose magnetic field is produced by an electric current ✓, and it can be turned on and off ✓.

b. Describe how you can make a simple electromagnet. (4)

To make a simple electromagnet, wrap a coil of wire around an iron nail ✓ and connect the ends of the wire to a battery. ✓ When the electric current flows through the wire, ✓ it creates a magnetic field around the nail.✓

visit twinkl.co.za

3.4. **Electrolysis**

a. Define electrolysis and give one example of its application. (2)

Electrolysis is the process of using an electric current to drive a non-spontaneous chemical reaction. ✓ An example is the electrolysis of water to produce hydrogen and oxygen gases. ✓

b. What happens to ions during the process of electrolysis? (2)

During electrolysis, ions move towards electrodes where they gain or lose electrons. ✓ Cations move to the cathode (negative electrode) and anions move to the anode (positive electrode). ✓

3.5. **Series and Parallel Circuits**

a. Draw a simple series circuit with three bulbs and one cell. Label all the parts. (4)

(4 marks) correct symbols of 3 bulbs and one cell

b. Draw a simple parallel circuit with three bulbs and one cell. Label all the parts. (4)

(4 marks) correct symbols of 3 bulbs (must be connected in parallel) and one cell

Section C

Question 4: Long Answer Questions **40**

4.1. **Safety with Electricity**

a. Explain the importance of safety measures when working with electricity. Provide three examples of safety measures. (4)

Safety measures are crucial when working with electricity to prevent accidents, injuries and damage to property. Examples include:

- **always turning off the power before working on electrical circuits**
- **using insulated tools and wearing rubber-soled shoes**
- **avoiding the use of electrical devices near water**

b. Describe the function of a fuse and how it protects electrical appliances. (4)

A fuse is a safety device that contains a thin metal strip which melts ✓ if the current flowing through the circuit exceeds a safe level. ✓ This interrupts the circuit ✓ and prevents damage to appliances and the risk of fire. ✓

visit twinkl.co.za

4.2. **Practical Application of Electromagnetism**

a. Discuss how electromagnets are used in everyday life. Provide at least two examples. (4)

Electromagnets are widely used in various applications:

In electric motors, where the electromagnet interacts with permanent magnets to create motion. √√

In magnetic levitation trains, where electromagnets are used to lift and propel the train. √√

b. Explain the relationship between electricity and magnetism in the context of electromagnets. (4)

Electromagnets work on the principle that an electric current produces a magnetic field √. When an electric current flows through a wire, it creates a magnetic field around the wire √. By coiling the wire and adding an iron core, the magnetic field is strengthened, √ creating a powerful electromagnet. √

4.3. **Chemical Effects of Electric Current**

a. Describe the process of electroplating and its benefits. (4)

Electroplating is the process of using an electric current √ to coat a metal object with a thin layer of another metal. √ This process is used to improve the appearance, corrosion resistance and hardness of the object √. For example, electroplating is used to coat jewellery with gold or silver. √

Total Marks: 70

visit twinkl.co.za